Fan-out two and exponential characteristic data: reductions, mobility lemmas, a universal obstruction for the doubling family, and polynomial data for abelian-filtered targets

v79 research draft for #2 (fixed-h MCFG), Open Problem `op:fanout-two`

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Abstract

We record five results related to Open Problem `op:fanout-two` of the v78 manuscript. (A) For every fan-out f, deleting one h-type from a target of $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ stays inside $\mathcal{C}^{\mathrm{mcf}}_{f,h}$; hence any target with a *rare type* (all of whose words are long) forces long characteristic data for the capped and uncapped reconstruction operators and for every set-driven operator learning both targets. (B) At fan-out two, a single long power u^K in one word of a target forces the target to be closed under transferring u-quanta between arbitrary u-blocks; more generally every sufficiently long word contains a constant-size interchangeable shift pair. (C) Consequently, the bracketed doubling family of the companion CFG paper is outside $\mathcal{C}^{\mathrm{mcf}}_{2,h}$ for *every* finite typing h once $n \geq 2|M|+3$, strengthening the fixed-window obstruction of v78. (D) For every finite typing h—including typings outside **LG**—and every fixed fan-out and rank, targets of the form $h^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathcal{H})$ with φ into an arbitrary abelian group admit characteristic samples of total length polynomial in grammar size and ordinary thickness. Thus the abelian-filter mechanism that produces most known members of $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ cannot yield a positive answer to `op:fanout-two`. The open problem itself remains open.

1 Setting

We use the definitions of the v78 manuscript verbatim: sentence contexts of arity d have nonempty internal separators; tuples have nonempty components; $\mathcal{D}^{(d)}_L(\vec{x})$ is the set of arity-d sentence contexts accepting $\vec{x}$; L is (f,h)-tuple-substitutable when, for $d \leq f$, equal componentwise h-types together with one shared accepting context imply equal distributions; $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ is the class of (f,h)-tuple-substitutable f-MCFLs; $\mathcal{B}^{\leq r}_{f,h}$ and $\mathcal{B}_{f,h}$ are the capped and uncapped reconstruction operators; $\|K\|_+ = \sum_{w\in K}(|w|+1)$; θ_G is rule thickness and t_G nonterminal thickness, with $t_G \leq \theta_G$ for reduced good G. For $L \subseteq \Sigma^*$ and $m \in M$ put

$$\mu_m(L) := \min\{|w| : w \in L,\ h(w) = m\} \qquad (\min \emptyset = \infty).$$

2 (A) Type deletion and the distinguishing pair

Lemma 1 (Type deletion)**.** *Let $f \geq 1$, $L \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$ and $m \in M$. Then $L^{\setminus m} := L \setminus h^{-1}(m) \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$.*

Proof. $h^{-1}(M \setminus \{m\})$ is regular, so $L^{\setminus m}$ is an f-MCFL (closure of f-MCFLs under intersection with regular languages, Seki et al. 1991). The language $h^{-1}(M \setminus \{m\})$ is (f,h)-tuple-substitutable (equal component types give the same total type in every context), and intersections of (f,h)-tuple-substitutable languages are (f,h)-tuple-substitutable (v78, Proposition *Intersection and abelian filtering*(i)). □

Proposition 2 (Rare types force long characteristic data). *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$ and $m \in h(L)$.*

(i) If $K \subseteq L$ is finite and contains no word of type m, then $L(\mathcal{B}^{\leq r}_{f,h}(K)) \subseteq L^{\setminus m} \subsetneq L$ for every $r \geq 1$, and likewise for $\mathcal{B}_{f,h}$. Hence every characteristic sample $\mathcal{F}$ of L for either operator contains a word of type m and $\|\mathcal{F}\|_+ \geq \mu_m(L) + 1$.

(ii) Let Op be any set-driven operator that has a characteristic sample for $L^{\setminus m}$. Then every characteristic sample of L for Op contains a word of type m.

Proof. (i) $K \subseteq L^{\setminus m} \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$, so soundness of every capped hypothesis (v78, Proposition *Soundness*) applied to the target $L^{\setminus m}$ gives the inclusion; $L \cap h^{-1}(m) \neq \emptyset$ gives strictness. (ii) If $\mathcal{F}$ is characteristic for L and avoids type m, and $\mathcal{F}'$ is characteristic for $L^{\setminus m}$, then $\mathcal{F} \cup \mathcal{F}' \subseteq L^{\setminus m} \subseteq L$ forces $\mathsf{Op}(\mathcal{F} \cup \mathcal{F}')$ to generate both L and $L^{\setminus m}$. □

Corollary 3 (Sufficient condition for `op:fanout-two`). *Suppose a fixed h and reduced good fan-out-$\leq f$, rank-$\leq r$ grammars G_n satisfy $L(G_n) \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$, $\|G_n\| \max\{1, \theta_{G_n}\} = n^{O(1)}$, and $\mu_{m_n}(L(G_n)) = n^{\omega(1)}$ for some types m_n. Then `op:fanout-two` has a positive answer, already against every set-driven operator with characteristic samples for the targets $L(G_n)$ and $L(G_n)^{\setminus m_n}$.*

This is exactly the mechanism of the companion CFG paper at fan-out one. Proposition 2 is not claimed to be necessary for a lower bound: a characteristic sample might be forced to be long by a sub-target $L' \subsetneq L$ in $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ that is not a type deletion.

3 (B) Mobility at fan-out two

Lemma 4 (Power transfer). *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$ with $f \geq 2$, let $u \in \Sigma^+$ and $g := h(u)$. Let $i_0 \geq 1$ and $\pi \geq 1$ satisfy $g^i = g^{i+\pi}$ for all $i \geq i_0$. Let $s, r \geq i_0$ and $d \in \pi\mathbb{Z}_{>0}$. If some word of L contains the factor $u^{s+d+r+2}$, then for all $X, Z \in \Sigma^*$ and $Y \in \Sigma^+$,*

$$X\,u^s\,Y\,u^{d+r}\,Z \in L \iff X\,u^{s+d}\,Y\,u^r\,Z \in L.$$

Proof. Write the given word as $Pu^{s+d+r+2}Q$. Put $\vec{x} = (u^s, u^{d+r})$, $\vec{y} = (u^{s+d}, u^r)$ and $E = (Pu)\,\Box_1\,u\,\Box_2\,Q$. All components are nonempty because $s, r \geq 1$, and E has the nonempty separator u. Then $E[\vec{x}] = E[\vec{y}] = Pu^{s+d+r+2}Q \in L$. Since $s, r \geq i_0$ and $\pi \mid d$, $h(u^s) = h(u^{s+d})$ and $h(u^{d+r}) = h(u^r)$. Arity-two substitutability gives $\mathcal{D}_L(\vec{x}) = \mathcal{D}_L(\vec{y})$; the displayed equivalence is membership of the context $X\Box_1 Y \Box_2 Z$. □

Remark 5. If g is the identity (e.g. u is a letter with trivial type), then $i_0 = \pi = 1$ and Lemma 4 applies as soon as some word contains u^5: one copy of u may then be moved between any two u-blocks of the indicated sizes, across arbitrary nonempty material. For a finite monoid, $i_0 + \pi \leq |M| + 1$, so $u^{2|M|+3}$ always suffices.

Lemma 6 (Local shift pairs). *Fix Σ and $h : \Sigma^* \to M$, let $c := |M|^2|\Sigma|$, and let $R := R_c(5)$ be the c-colour Ramsey number for monochromatic K_5. Every $w \in \Sigma^*$ with $|w| \geq R$ has a factorisation*

$$w = P\,x_1\,p\,q\,p\,y_2\,Q, \qquad x_1, p, y_2 \in \Sigma^+,\ q \in \Sigma^*,$$

with $h(x_1) = h(x_1pq)$, $h(qpy_2) = h(y_2)$ and $|p| = 1$. Consequently, whenever $L \in \mathcal{C}^{\mathrm{mcf}}_{2,h}$ contains w, the tuples $\vec{x} = (x_1, qpy_2)$ and $\vec{y} = (x_1pq, y_2)$ satisfy $\mathcal{D}_L(\vec{x}) = \mathcal{D}_L(\vec{y})$, i.e. for all X, Z and $Y \neq \varepsilon$: $Xx_1Yqpy_2Z \in L \iff Xx_1pqYy_2Z \in L$.

Proof. Colour each pair $i < j$ of positions in $\{0, \dots, |w|-1\}$ by $\big(h(w[i{:}j]),\, h(w[i{+}1{:}j]),\, w[i]\big)$, where $w[i{:}j] = w_{i+1} \cdots w_j$ in the cut-position numbering of v78 and $w[i]$ is the letter after cut i. By Ramsey's theorem there are $i_1 < \cdots < i_5$ with all ten pairs of one colour. Put $x_1 = w[i_1{:}i_2]$, $p = w[i_2] = w[i_3]$ (third coordinates of (i_2, i_5) and (i_3, i_5)), $q = w[i_2{+}1{:}i_3]$ and $y_2 = w[i_3{+}1{:}i_5]$, which is nonempty since $i_5 \geq i_3 + 2$. The first coordinates of $(i_1, i_2), (i_1, i_3)$ give $h(x_1) = h(x_1pq)$, and the second coordinates of $(i_2, i_5), (i_3, i_5)$ give $h(qpy_2) = h(y_2)$. The context $E = P\Box_1 p \Box_2 Q$ with $P = w[0{:}i_1]$, $Q = w[i_5{:}|w|]$ has the nonempty separator p and $E[\vec{x}] = E[\vec{y}] = w$. □

Lemma 6 says that at fan-out two every long word of a target carries an interchangeable pair of bounded size whose components may then be transported to *any* other occurrence in the language. This local mobility has no fan-out-one analogue: an arity-one context $P\Box Q$ determines its filling.

4 (C) No finite typing rescues the doubling family

Let $\Gamma = \{\mathtt{a}, \lfloor, \rfloor, \lceil, \rceil\}$, $W_0 = \mathtt{a}$, $W_i = \lfloor W_{i-1} W_{i-1} \rfloor$, and let L_n be generated from E_n by $E_0 \to \mathtt{a}$, $E_i \to \lfloor E_{i-1} E_{i-1} \rfloor \mid \lceil E_{i-1} \rceil$. The unique word of L_n without ceiling brackets is $W_n = \lfloor^n \mathtt{a} \cdots$.

Theorem 7. *Let $h : \Gamma^* \to M$ be any finite-monoid homomorphism, and let $i_0 \geq 1, \pi \geq 1$ be the index and period of $g = h(\lfloor)$ on positive powers. For every $n \geq 2i_0 + \pi + 2$, in particular for every $n \geq 2|M| + 3$, $L_n \notin \mathcal{C}^{\mathrm{mcf}}_{2,h}$ (hence $L_n \notin \mathcal{C}^{\mathrm{mcf}}_{f,h}$ for all $f \geq 2$).*

Proof. Suppose $L_n \in \mathcal{C}^{\mathrm{mcf}}_{2,h}$. Put $s = i_0$, $d = \pi$, $r = n - s - d - 2 \geq i_0$. The word W_n begins with $\lfloor^n$, so Lemma 4 applies with $u = \lfloor$. Write $W_n = \lfloor\, W_{n-1} W_{n-1} \rfloor$; the second copy of W_{n-1} begins with $\lfloor^{n-1}\mathtt{a}$ and $d + r = n - s - 2 \leq n - 1$. Choose the factorisation $W_n = X \lfloor^s Y \lfloor^{d+r} Z$ with $X = \lfloor$ (so that $\lfloor^s$ occupies positions $1..s$ of the leading run) and with $\lfloor^{d+r}$ the last $d + r$ floors of the leading run of the second copy of W_{n-1}; Y is nonempty because it contains the first $\mathtt{a}$. The lemma gives $W' := X \lfloor^{s+d} Y \lfloor^r Z \in L_n$. But W' has no ceiling bracket and its first $\mathtt{a}$ is preceded by $n + d \neq n$ floors, so $W' \neq W_n$, a contradiction. Finally $i_0 + \pi \leq |M| + 1$ and $i_0 \leq |M|$ give $2i_0 + \pi + 2 \leq 2|M| + 3$. □

Remark 8. v78 proved the special cases $h = h^{k,\ell}_\Gamma$ and $h = (h^{k,\ell}_\Gamma, h_{\mathrm{U}})$ with the sharper threshold $n \geq \max\{3, k{+}\ell{+}1\}$. Theorem 7 has a cruder threshold but quantifies over *all* finite typings, so no typing at all—windows, groups, semilattices, or any product—repairs this family at fan-out two. The obstruction is the long run $\lfloor^n$, not the choice of h.

5 (R) Relaxed witnesses: what the data length really measures

For a reduced good presentation G of $L \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$ let $\widetilde{G}_0$ be its trimmed yield-type refinement (v78). For a non-start typed nonterminal $X = A_{\mathbf{t}}$ write $[X]$ for the common $\equiv_{L,h}$-class of the tuples in L_X (it is one class: equal type $\mathbf{t}$ and a shared canonical context, v78), and put

$$\kappa_G(X) := \min\{\|\vec{v}\|_1 : \vec{v} \in [X]\}, \qquad \kappa_G := \max_X \kappa_G(X).$$

Lemma 9 (Relaxed witnesses)**.** *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$ and let G be a reduced good presentation of fan-out $\leq f$ and rank $\leq r$. For every non-start typed nonterminal $X = A_{\mathbf{t}}$ choose any $\omega'(X) \in [X]$, and let $\chi'(X)$ be $\Box_1$ if $\widetilde{S} \to X$ is a rule of $\widetilde{G}_0$, and otherwise any derivational sentence context of the untyped nonterminal A in G. Let $K^\star$ consist of the words $\chi'(X)[\omega'(X)]$, $\chi'(X)[\rho(\omega'(Y_1), \dots, \omega'(Y_q))]$ for non-start typed rules $X \to \rho(Y_1, \dots, Y_q)$, and $\chi'(X)[\vec{c}]$ for non-start rank-zero typed rules $X \to \vec{c}$. Then $K^\star \subseteq L$ and $K^\star$ is a characteristic sample of L for $\mathcal{B}^{\leq r}_{f,h}$ and for $\mathcal{B}_{f,h}$.*

Proof. A derivational context of A accepts every tuple of $L_A(G) \supseteq L_X$, hence every tuple of $[X]$; it lies in $\mathcal{E}_{\mathrm{fan}(A)}$ by goodness (v78, Lemma *Canonical contexts*). If $\widetilde{S} \to X$ is present, every tuple of L_X is a one-component word of L, so $\Box_1 \in \mathcal{D}_L(\omega'(X))$. Thus $\chi'(X)[\omega'(X)] \in L$. For a typed rule, $\omega'(Y_j) \equiv_{L,h} \vec{u}_j$ for $\vec{u}_j \in L_{Y_j}$, so v78, Lemma *Composition preserves equivalence*, gives $\rho(\omega'(Y_1), \ldots) \equiv_{L,h} \rho(\vec{u}_1, \ldots) \in L_X$; rank-zero rules produce members of L_X. Hence all words of $K^\star$ lie in L, and $\omega'(X)$ has the same type as every rule tuple of X and shares the context $\chi'(X)$ with it. These are the only properties used by the completeness proof of v78 (Proposition *Completeness from a target presentation* and Lemma *Target rule words occur in the enumeration*): tuple components are nonempty, templates are good and $\chi'(X) \in \mathcal{E}_d$, so the child tuples are observed occurrences. Completeness and soundness give the claim. □

Corollary 10 (Upper bound by class representatives)**.** *In the situation of Lemma 9 one may take*

$$|K^\star| \le |V||M|^f + |P||M|^{fr}, \qquad \|K^\star\|_+ \le \big(|V||M|^f + |P||M|^{fr}\big)\big(1 + \theta_G + \|G\| + (r+1)\kappa_G\big).$$

Consequently, for a fixed typing and fixed f, r, a positive answer to `op:fanout-two` *for a family* G_n requires $\kappa_{G_n} = n^{\omega(1)}$*: some typed distribution class of G_n has only superpolynomially long members.*

Proof. A minimal successful derivation using a rule with left-hand side A gives a derivational context of fixed length at most θ_G. Each selected word consists of such a context, at most $\|G\|$ explicit terminals, and at most r tuples of total length at most κ_G (choosing $\omega'(X)$ of minimum length). □

Remark 11. Proposition 2 is the special case of a long class at a start-linked typed nonterminal $S_{(m)}$: its class consists of words of L of type m, so $\kappa_G(S_{(m)}) \ge \mu_m(L)$. The exponential *full-refinement* thickness of v78 (Proposition *Exponential blow-up of full yield-type refinement*) does not by itself produce long data: there $[X]$ contains short tuples although L_X does not.

6 (D) Abelian-filtered targets have polynomial data

Theorem 12. *Fix finite Σ, a finite typing $h : \Sigma^* \to M$, $f \ge 1$ and $r \ge 1$. There are constants C, e depending only on Σ, h, f, r such that the following holds. Let $\mathbb{A}$ be an abelian group, $\varphi : \Sigma^* \to \mathbb{A}$ a homomorphism, $\mathcal{H} \le \mathbb{A}$, $\gamma \in \mathbb{A}$, $\mathsf{Acc} \subseteq M$, and suppose*

$$L := h^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathcal{H})$$

has a reduced good presentation $G = (V, \Sigma, P, S)$ of fan-out $\le f$ and rule rank $\le r$. Then $L \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$, and $\mathcal{B}^{\le r}_{f,h}$ has a characteristic sample $K^\star \subseteq L$ with

$$|K^\star| \le |V||M|^f + |P||M|^{fr}, \qquad \|K^\star\|_+ \le C\left(\|G\| \max\{1, \theta_G\}\right)^e.$$

In particular $\mu_m(L) \le C(\|G\| \max\{1, \theta_G\})^e$ for every $m \in h(L)$.

The proof uses two external facts. *Parikh's theorem*: for each $m \in M$ the Parikh images of $h^{-1}(m)$ and of $h^{-1}(m) \cap \Sigma^+$ are semilinear; we fix such representations once and for all, so their number of linear sets, their constants and their periods are bounded by a constant C_h depending only on h. *Small solutions of integer programs* (Papadimitriou, J. ACM 28 (1981) 765–768): if $Az = b$, $z \ge 0$ has an integer solution, where A has m rows, n columns and all entries of A, b lie in $[-a, a]$, then it has one with all entries at most $n(ma)^{2m+1}$.

Proof. Membership. L is an f-MCFL by hypothesis. The language $h^{-1}(\mathsf{Acc})$ is (f,h)-tuple-substitutable, $\varphi^{-1}(\gamma+\mathcal{H})$ is (f,h_{triv})-tuple-substitutable for an arbitrary abelian group $\mathbb{A}$, and intersections preserve substitutability (v78, Proposition *Intersection and abelian filtering*); hence $L \in \mathcal{C}^{\mathrm{mcf}}_{f,h}$.

Notation. Let $\psi : \Sigma^* \to \mathbb{N}^\Sigma$ be the Parikh map and, for a tuple, $\Psi(\vec{u}) = \sum_i \psi(u_i)$. Since $\mathbb{A}$ is abelian, $\varphi = \bar{\varphi} \circ \psi$ for a group homomorphism $\bar{\varphi} : \mathbb{Z}^\Sigma \to \mathbb{A}$. Let $\Lambda := \{v \in \mathbb{Z}^\Sigma : \bar{\varphi}(v) \in \mathcal{H}\}$. For a word or tuple $\vec{u}$ and a sentence context E, $\psi(E[\vec{u}]) = \psi(E) + \Psi(\vec{u})$, where $\psi(E)$ counts the fixed material.

Step 1 (value invariance). For $A \in V$ and $\vec{u}, \vec{u}' \in L_A(G)$, $\Psi(\vec{u}) - \Psi(\vec{u}') \in \Lambda$. Indeed A occurs in a successful derivation; removing the subtree gives a derivational context E with $E[\vec{u}], E[\vec{u}'] \in L$ (nondeleting linear templates), and both values lie in $\gamma + \mathcal{H}$.

Step 2 (small generators). For $A \in V$ fix $\omega_A \in L_A(G)$ with $\|\omega_A\|_1 \le t_G$. For a rule $R : A \to \rho(B_1, \dots, B_q)$ let $c_R \in \mathbb{N}^\Sigma$ count its terminals and put $\delta_R := \Psi(\omega_A) - c_R - \sum_j \Psi(\omega_{B_j})$. Since $\rho(\omega_{B_1}, \dots, \omega_{B_q}) \in L_A(G)$, Step 1 gives $\delta_R \in \Lambda$, and $\|\delta_R\|_1 \le T := (r+1)t_G + \|G\|$. Let $\Gamma_T := \{\lambda \in \Lambda : \|\lambda\|_1 \le T\}$ and let Λ_T be the subgroup it generates; $|\Gamma_T| \le (2T+1)^{|\Sigma|}$.

Step 3. For every $\vec{u} \in L_A(G)$, $\Psi(\vec{u}) - \Psi(\omega_A) \in \Lambda_T$, by induction on a derivation: if the root rule is R with child yields $\vec{u}_j$, then $\Psi(\vec{u}) - \Psi(\omega_A) = \sum_j (\Psi(\vec{u}_j) - \Psi(\omega_{B_j})) - \delta_R$.

Step 4 (short typed tuples). Let $X = A_{\mathbf{t}}$ be a non-start typed nonterminal of the trimmed refinement $\widetilde{G}_0$, $d = \mathrm{fan}(A)$, and fix $\vec{u} \in L_X$. For each $i \le d$ choose a linear set $\{c_i + \sum_{p \in P_i} \nu_p p\}$ of the fixed representation of $\psi(h^{-1}(t_i) \cap \Sigma^+)$ containing $\psi(u_i)$. The system

$$\sum_i \Big(c_i + \sum_{p \in P_i} \nu_{i,p}\, p \Big) - \sum_{\lambda \in \Gamma_T} (\mu^+_\lambda - \mu^-_\lambda)\lambda = \Psi(\omega_A), \qquad \nu, \mu^\pm \ge 0 \text{ integers},$$

has $|\Sigma|$ equations and is solvable by Step 3 applied to $\vec{u}$. Its entries are bounded by $a := T + fC_h$, and it has at most $fC_h + 2(2T+1)^{|\Sigma|}$ unknowns. By Papadimitriou's bound it has a solution with all entries $\le \mathrm{poly}(T)$, the degree depending only on $|\Sigma|$. Each vector $c_i + \sum \nu_{i,p} p$ is the Parikh vector of some $\omega'_i \in h^{-1}(t_i) \cap \Sigma^+$. Put $\omega'(X) := (\omega'_1, \dots, \omega'_d)$; then $h^{(d)}(\omega'(X)) = \mathbf{t}$, $\Psi(\omega'(X)) - \Psi(\omega_A) \in \Lambda_T$, and $\|\omega'(X)\|_1 \le \mathrm{poly}(T)$.

Step 5 (equivalence). If two tuples $\vec{v}, \vec{v}'$ have the same componentwise type and $\Psi(\vec{v}) - \Psi(\vec{v}') \in \Lambda$, then $E[\vec{v}] \in L \iff E[\vec{v}'] \in L$ for every context E: the total types coincide and $\varphi(E[\vec{v}]) - \varphi(E[\vec{v}']) = \bar{\varphi}(\Psi(\vec{v}) - \Psi(\vec{v}')) \in \mathcal{H}$.

Step 6 (conclusion via relaxed witnesses). By Step 5 and Step 4, $\omega'(X) \equiv_{L,h} \vec{u}$ for every $\vec{u} \in L_X$, because $\Psi(\omega'(X)) - \Psi(\vec{u}) = (\Psi(\omega'(X)) - \Psi(\omega_A)) - (\Psi(\vec{u}) - \Psi(\omega_A)) \in \Lambda_T \subseteq \Lambda$. Apply Lemma 9 with these $\omega'(X)$. By Corollary 10,

$$\|K^\star\|_+ \le \big(|V||M|^f + |P||M|^{fr}\big)\big(1 + \theta_G + \|G\| + (r+1)\max_X \|\omega'(X)\|_1\big),$$

and $\max_X \|\omega'(X)\|_1 \le \mathrm{poly}(T)$ with $T \le (r+1)\theta_G + \|G\|$ because $t_G \le \theta_G$. Finally, by Proposition 2(i) the characteristic sample $K^\star$ contains a word of every type $m \in h(L)$, whence $\mu_m(L) \le \|K^\star\|_+$. □

Remark 13. Theorem 12 holds for typings outside **LG**, including h_{U}, and at every fan-out, including fan-out one. The bracketed CFG family of the companion paper is not abelian-filtered. Since the abelian-filter construction is the mechanism behind the members of $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ exhibited in v78 (e.g. MIX^+_2 and L_{intr}), any positive answer to `op:fanout-two` must use targets whose tuple distributions are not determined by type and an abelian invariant. A finite union of abelian filters on type-disjoint branches (such as $L_\star$) is expected to admit the same argument branch by branch; this is stated as a conjecture in the memo, not proved here.

7 Conjectures

Conjecture 14 (Short class representatives at fan-out two)**.** For every finite Σ, finite typing h and $r \geq 1$ there is a polynomial p such that every reduced good presentation G of fan-out ≤ 2 and rank $\leq r$ with $L(G) \in \mathcal{C}^{\mathrm{mcf}}_{2,h}$ satisfies $\kappa_G \leq p(\|G\| \max\{1, \theta_G\})$.

By Corollary 10, Conjecture 14 *implies* a negative answer to `op:fanout-two` at $f = 2$ for the operators $\mathcal{B}^{\leq r}_{2,h}$, for every fixed h (the converse is not claimed). It is false at fan-out one (companion CFG paper), true for $h(\Sigma^+) \in \mathbf{LG}$ (v78) and for abelian-filtered targets (Theorem 12).

Conjecture 15 (Rare types are polynomially short at fan-out two)**.** For every finite Σ, finite typing h and $r \geq 1$ there is a polynomial p such that, for every reduced good presentation G of fan-out ≤ 2 and rank $\leq r$ with $L(G) \in \mathcal{C}^{\mathrm{mcf}}_{2,h}$ and every $m \in h(L(G))$, $\mu_m(L(G)) \leq p(\|G\| \max\{1, \theta_G\})$.

The conjecture holds for h with $h(\Sigma^+) \in \mathbf{LG}$ (v78) and for abelian-filtered targets (Theorem 12); it fails at fan-out one (companion CFG paper). If it holds, the type-deletion method of Proposition 2 cannot settle `op:fanout-two`.